Experiment 4: The Contrastive Adjustment $d = E - A$ Does Not Suppress Hallucinations, and a Noise Proxy Reproduces Its Effect

d-analysis and magnitude-matched noise controls on LLaVA-1.5-7B and Qwen2.5-VL-7B

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1 Hypothesis

Contrastive decoding, in the visual forms VCD [Leng et al., 2024] and SID [Huo et al., 2024], builds its output score as $C = (1+\alpha)E - \alpha A$, which at $\alpha = 1$ equals $2E - A = E + d$, where $d = E - A$ is the gap between the clean-image expert logit E and the degraded-image amateur logit A. The quantity d is therefore exactly the adjustment that contrastive decoding adds to the plain expert score at every candidate token. The stated purpose of the method is to push down tokens that the language prior favours without visual support, so for a hallucinated object d should be negative.

We test this premise in two complementary ways.

H1 (Selective suppression). If contrastive decoding suppresses hallucinations, the adjustment d should be more negative for hallucinated objects than for real objects. We measure the sign and size of d separately for the two groups. A working suppressor predicts clearly negative d on hallucinations, and more negative on hallucinations than on real objects.

H2 (Mechanism versus perturbation). If the change contrastive decoding produces in the hallucination rate is a targeted effect of the amateur branch, then replacing that branch with random noise of the same magnitude should not reproduce it. We construct noise proxies whose only input is expert logits plus a random bonus matched to the measured size of d, and compare their CHAIR scores against the real methods.

The first test asks whether the contrastive adjustment points in the direction the method claims. The second asks whether the measured effect on hallucination requires the amateur branch at all, or whether generic noise of the same magnitude is enough to explain it. We run both on two model families (LLaVA-1.5-7B [Liu et al., 2024] and Qwen2.5-VL-7B [Bai et al., 2025]) and both contrastive methods (VCD and SID). This complements the benchmark-level critique of Yin et al. [2025] by examining the logit adjustment itself.

2 Methodology

The analysis has two parts that use the same 500 images but not the same decode. Part one (Section 2.2) reads the contrastive adjustment $d = E - A$ from logits stored along the real VCD or SID decoding trajectory (the path chosen by $C = 2E - A$). Part two (Section 2.3) performs an independent expert-only decode with random noise added to the logits, and asks whether that noise stand-in reproduces the real methods' effect on hallucination. Only the image set is shared; the two parts are separate decoding runs. We first fix the common setup.

2.1 Shared setup

Models and data. We use LLaVA-1.5-7B and Qwen2.5-VL-7B-Instruct, both in `float16` with `sdpa` attention, on a fixed set of 500 MSCOCO [Lin et al., 2014] `val2017` images (the same images and order for every run, so all comparisons are paired). The prompt is "Describe this image in detail" and decoding is greedy with `max_new_tokens` $= 256$.

Contrastive branches. The expert branch E scores the next token from the clean image. The amateur branch A scores it from a degraded image. For VCD the degradation is forward diffusion noise at step 500; for SID it is attention pruning that keeps a random 12.5% of image tokens from layer two onward. The contrastive score is $C = 2E - A$ (that is $\alpha = 1$), followed by the adaptive plausibility constraint (APC), which removes any token whose expert logit is below $\log \beta + \max_w E_t(w)$ with $\beta = 0.2$. These settings match the reference implementations and the other experiments in this paper.

What is recorded. At every decoding step we store the top-30 token ids and logits of both the expert and the amateur branch, together with the token the method finally emits. All object labelling is done afterward from the generated caption text, never from raw sub-word tokens, which avoids the sub-word mislabelling problem (for example the "ship" inside "companionship" or a "bear" split off from "teddy bear").

Object labelling. We use the standard CHAIR protocol [Rohrbach et al., 2018]. Each image has a ground-truth object set built from its COCO instance-segmentation labels plus the objects named in its five human reference captions, mapped to the 80 MSCOCO categories through the CHAIR synonym list. An object the model mentions is *real* if its category is in that set and *hallucinated* if it is not.

2.2 Part one: the contrastive adjustment $d = E - A$

For a given token we read its expert logit E and amateur logit A from the stored top-30 lists and form $d = E - A$. Because $C = E + d$, a positive d means the token is amplified by contrastive decoding and a negative d means it is suppressed. We compute d over three populations of object tokens, from the cleanest to the broadest, so that the conclusion does not rest on one arbitrary cut-off.

1. **Emitted object words (primary).** These are the object words the model actually wrote, labelled at the caption level, so every real or hallucinated tag is exact. For each such word we take the step that produced its first token and read E and A for that token.
2. **Top-10 expert candidates (robustness).** At every step we take the ten highest tokens by expert logit, keep those that are word-initial COCO object words, and label each by ground-truth membership. This is not conditioned on a word winning, so it removes any selection effect of the emitted set.
3. **Top-30 expert candidates (robustness).** The same as above with a wider window, to check that the picture is stable as the candidate set grows.

For each population we report, separately for real and hallucinated objects, the mean of d and the fraction of tokens with $d > 0$ (the share that contrastive decoding amplifies rather than suppresses). We visualise the full spread with per-token bar plots, one bar per token, coloured red where $d > 0$ and blue where $d < 0$, with the group mean drawn as a dashed line.

Two honest notes. First, if a token is not present in both the expert and the amateur stored top-30 lists, its d cannot be formed and the token is skipped; this is a small fraction and is applied identically to both branches, so it cannot bias the real-versus-hallucinated comparison. Second, the candidate populations use single-token labels, so a word-initial "bear" that begins "teddy bear" would be read as the animal; this affects well under one percent of candidates and hits both branches equally. The emitted population has no such caveat, which is why we treat it as primary.

2.3 Part two: the noise proxy

The measurements in Section 2.2 suggest a specific reading of the contrastive term. The adjustment $d = E - A$ that contrastive decoding adds to the expert logits does not behave like a targeted, content-aware signal: it does not single out hallucinated objects for suppression, and across tokens

Table 1: Noise parameters for the proxies, taken from the measured distribution of $d = E - A$ over object tokens. Proxy A draws from a Gaussian with these mean and standard deviation; Proxy B resamples the empirical d values (same mean and standard deviation by construction).

Model	Method	mean of d	std of d
LLaVA-1.5-7B	VCD	$+0.199$	0.853
	SID	$+0.300$	0.903
Qwen2.5-VL-7B	VCD	-0.846	1.531
	SID	-0.663	1.558

it resembles a small, roughly zero-centred perturbation layered on top of the expert scores. This motivates a direct test. If the amateur branch contributes, in effect, only a perturbation of a certain size, then adding a matched amount of *random* noise to the same expert logits should produce a similar outcome, and the amateur branch would not be doing anything a generic perturbation could not.

We therefore build a deliberately minimal control, the noise proxy. We discard the amateur branch entirely. Starting from the plain expert logits E, we add independent random noise to the object-category logits, with magnitude matched to the measured distribution of d for the real method, and then apply the same adaptive plausibility constraint and greedy decoding. The proxy has no access to the degraded image and performs no contrast; its only ingredients are the expert logits and random noise of the same size that contrastive decoding actually adds. We evaluate it on the standard CHAIR benchmark under identical conditions and compare against the real VCD and SID runs. The logic is straightforward: if contrastive decoding worked through a genuine image-grounded mechanism, matched random noise should not reproduce its effect on hallucination; if the proxy lands at the same CHAIR scores as the real methods, the measured effect of contrastive decoding is statistically indistinguishable from adding random noise of the same size, and the amateur branch is not contributing targeted, hallucination-reducing information.

Object-token vocabulary. We first collect the set of vocabulary token ids that are word-initial and spell an MSCOCO object word (348 tokens for Qwen; a comparable set for LLaVA). These are the only tokens the proxy perturbs, because these are the tokens whose logits determine CHAIR. Restricting the noise to object tokens is deliberate and conservative: it gives the proxy the best possible chance to mimic a hallucination-relevant intervention.

Which d distribution the noise is matched to. The noise parameters in Table 1, and the empirical values that Proxy B resamples, are computed over a single broad pool: every object-category token that appears anywhere in the stored top-30 expert candidate list, at every step, with real and hallucinated tokens combined (about $114{,}000$ tokens per model). This pool is deliberately broader than the top-10, top-30, or emitted populations used in the d-analysis of Section 2.2, which is why its mean does not equal any single value in those tables. Using one pooled distribution keeps the proxy agnostic to whether a token is real or hallucinated, which is the point of the control.

Proxy A (Gaussian, magnitude-matched). We run expert-only greedy decoding. At each step we add to every object-token logit an independent draw from a Gaussian whose mean and standard deviation equal the measured pooled mean and standard deviation of d for the corresponding real method (Table 1). We then apply the same APC and take the argmax. There is no amateur branch and no second image pass.

Proxy B (empirical bootstrap). Identical to Proxy A, except the bonus for each object token is drawn by resampling with replacement from the empirical set of measured d values for that method. This reproduces the real distribution of d, including its skew, not just its first two moments.

Matching the magnitude. Table 1 lists the noise parameters. The sign of d differs by model: on LLaVA the mean of d is positive, so the proxy on average raises object logits; on Qwen it is negative, so the proxy on average lowers them. In both cases the proxy matches the real method's own measured magnitude, so it is a like-for-like stand-in.

Scoring. Every proxy run produces 500 captions on the same images, which we score with the standard CHAIR metric using the identical scorer applied to the real greedy, VCD, and SID runs. We report CHAIR-S (the percentage of captions containing at least one hallucinated object) and

Table 2: **Contrastive adjustment $d = E - A$ over the expert's top-10 object candidates.** Mean of d and the percentage of tokens with $d > 0$ (the share amplified rather than suppressed), for real and hallucinated objects.

		Real objects		Hallucinated objects	
Model	Method	mean d	% $d>0$	mean d	% $d>0$
LLaVA-1.5-7B	VCD	$+0.133$	56.9	$+0.164$	57.2
	SID	$+0.218$	59.3	$+0.114$	50.3
Qwen2.5-VL-7B	VCD	-0.984	25.1	-0.429	36.1
	SID	-0.674	31.9	-0.172	41.6

CHAIR-I (the percentage of mentioned object instances that are hallucinated). Lower is better for both.

Reproducibility note. No global random seed was fixed. The noise draws in the proxies and the stochastic degradations in the real methods therefore differ across runs, but every quantity is aggregated over 500 images and tens of thousands of tokens, a regime where this variation is negligible relative to the reported effects.

3 Results

3.1 The contrastive adjustment does not selectively suppress hallucinations

Table 2 reports $d = E - A$ over the expert's top-10 object candidates at each step. We use the candidate set rather than only the emitted word because it is not conditioned on a token winning, so it carries no selection effect. The finding is clearest on Qwen and weaker but consistent on LLaVA, and we state it precisely rather than as a blanket claim.

On Qwen the adjustment is negative overall, meaning the method lowers object logits, but it lowers *real* objects far more than hallucinations (VCD mean -0.984 for real versus -0.429 for hallucinated; SID -0.674 versus -0.172), and the share of amplified tokens is higher for hallucinations (36.1% versus 25.1% under VCD). This is the opposite of the intended behaviour and it holds at every window. On LLaVA the adjustment is small in magnitude for both groups and its ordering is not stable: under VCD hallucinated d is marginally above real ($+0.164$ versus $+0.133$), while under SID it is below real ($+0.114$ versus $+0.218$), with both groups amplified rather than suppressed. The LLaVA-SID row is the only one of the four in which hallucinated d sits below real d, which is the direction H1 predicts; we return to it below, because it does not survive the other two views.

Figures 1 and 2 show the full per-token spread over the top-10 candidates. Red bars are tokens contrastive decoding amplifies ($d > 0$), blue bars are tokens it suppresses ($d < 0$), and the dashed line is the group mean. On Qwen the real-object panels are markedly more negative than the hallucination panels, which is the visual form of the numbers above.

Robustness across windows. Table 3 widens the candidate set to the top-30. On Qwen the ordering is unchanged: hallucinated d stays well above real d. On LLaVA both means shrink toward zero, and here hallucinated d is marginally *below* real d for both methods (VCD -0.004 versus $+0.041$; SID -0.018 versus $+0.154$). These LLaVA gaps are tiny (all $|d| \leq 0.16$), and their sign is not stable: the LLaVA-SID case that looked mildly suppressive at top-10 and top-30 reverses completely in the exact-label emitted view below ($+0.684$ for hallucinated versus $+0.296$ for real). We therefore do not read the small LLaVA numbers as evidence of suppression; they are consistent with noise around zero.

Emitted words with exact labels. The candidate populations rely on single-token labels. As a co-primary check with exact caption-level labels, we repeat the measurement on the object words the model actually emits (Table 4, and Figures 3 and 4). Here the four settings agree in one direction: emitted hallucinated objects receive a *higher* adjustment than real objects in every case (for example LLaVA-SID $+0.684$ versus $+0.296$, and Qwen-SID $+0.429$ versus -0.747, where hallucinations are amplified while real objects are suppressed). We note honestly that for LLaVA-SID this exact-label view points the opposite way to the top-10 and top-30 candidate views: the candidate views put hallucinated d slightly below real, the emitted view puts it well above. The two views agree only

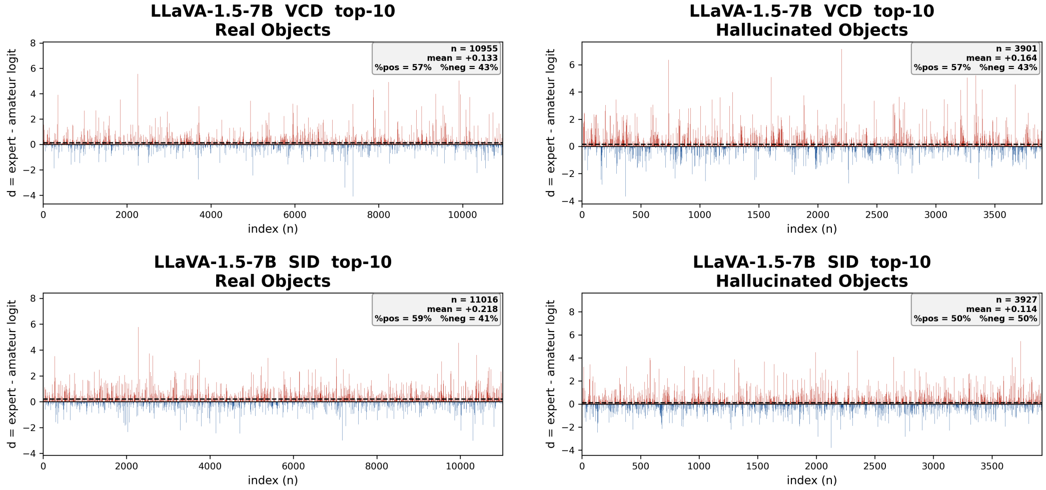

Figure 1: **LLaVA-1.5-7B: $d = E - A$ over top-10 object candidates.** Left column real objects, right column hallucinated objects; top row VCD, bottom row SID. The adjustment is positive (amplifying) and small for both groups.

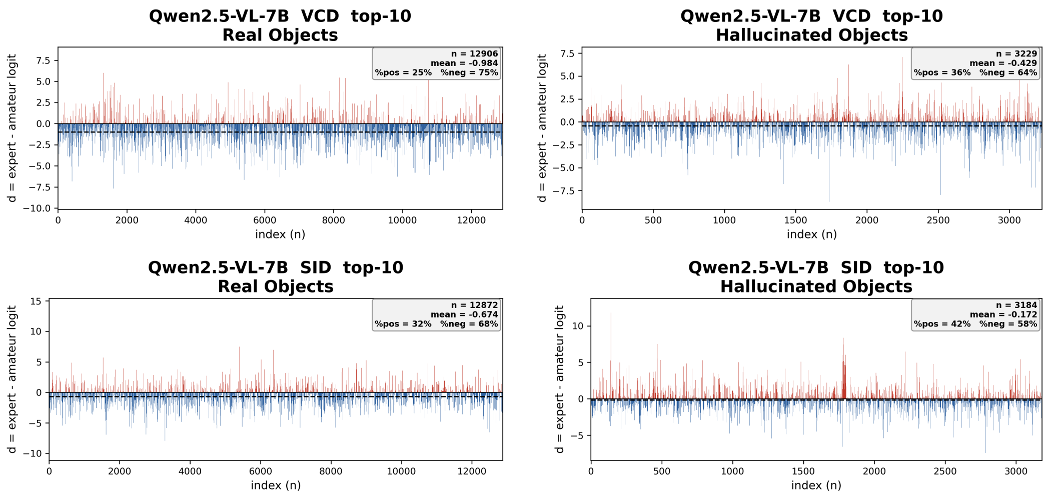

Figure 2: **Qwen2.5-VL-7B: $d = E - A$ over top-10 object candidates.** Layout as in Figure 1. Real objects are suppressed more than hallucinations, the opposite of the intended behaviour.

in the weak sense that neither shows the clean negative suppression H1 predicts; they do not agree on the sign of the real-versus- hallucinated gap for LLaVA-SID. This instability is itself the point: there is no stable suppressive signal to be found.

3.2 Random noise of matched magnitude matches the effect on LLaVA and is exceeded by the real method on Qwen

Table 5 reports CHAIR with image-level bootstrap 95% confidence intervals, so the proxy-versus-real comparison rests on intervals rather than single point estimates. All rows use one consistent set of runs (the same VCD and SID runs that feed the d-analysis above).

On **LLaVA**, the real methods and the noise proxies have heavily overlapping intervals on both metrics. Real VCD CHAIR-I is 15.6 $[14.1, 17.0]$ and the two VCD noise proxies are 14.8 $[13.3, 16.3]$ and 14.3 $[12.9, 15.8]$; real SID is 15.1 $[13.7, 16.6]$ and its proxies are 14.2 and 14.5. The difference between real contrastive decoding and magnitude-matched random noise is therefore within image-sampling variance. Neither real method separates from greedy either (greedy CHAIR-I 13.9

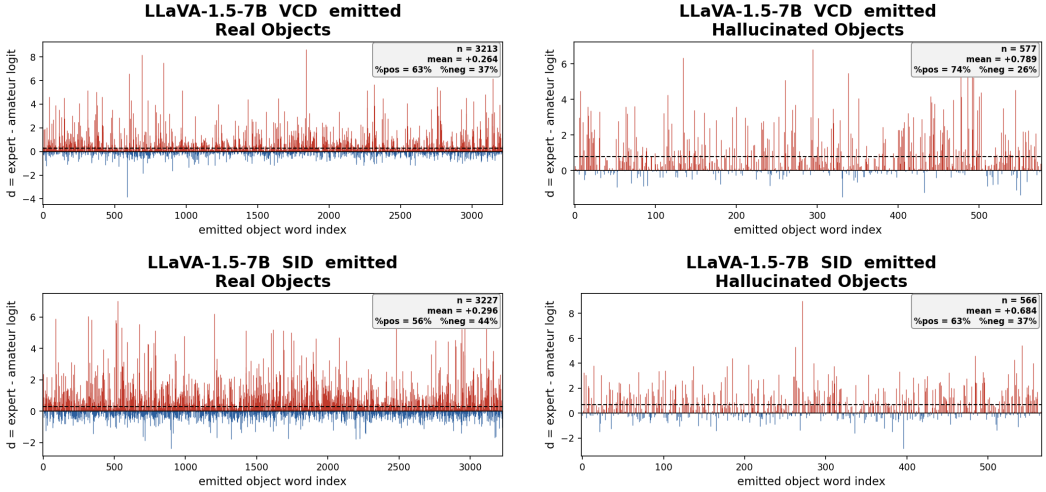

Figure 3: **LLaVA-1.5-7B:** $d = E - A$ **on emitted object words** (exact caption-level labels; one thin spike per emitted word). Left column real, right column hallucinated; top row VCD, bottom row SID. Hallucinated words are amplified more than real words.

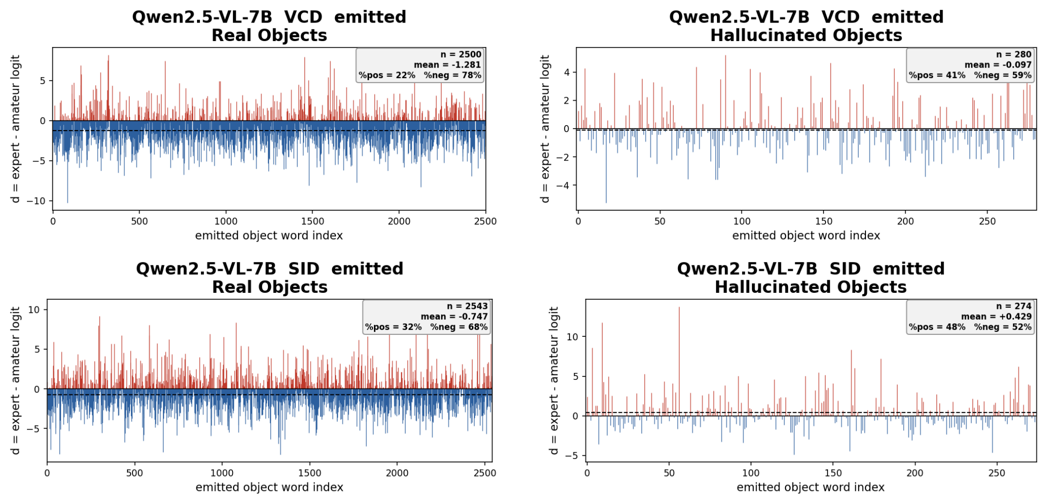

Figure 4: **Qwen2.5-VL-7B:** $d = E - A$ **on emitted object words** (exact caption-level labels; one thin spike per emitted word). Layout as in Figure 3. Real objects are strongly suppressed while hallucinations sit near zero or are amplified.

$[12.5, 15.3]$). In short, on LLaVA the amateur branch does nothing to CHAIR that random noise of the same size on object logits does not already do.

On **Qwen**, the picture is different and, for the method, worse. Real VCD and SID raise CHAIR-I to 10.5 $[8.9, 12.1]$ and 10.5 $[9.1, 12.0]$, above greedy at 7.7 $[6.5, 8.9]$, while the noise proxies stay near greedy at 8.2 $[6.9, 9.5]$ and 8.3 $[7.0, 9.6]$. Here matched noise does *not* reproduce the degradation: the real subtraction pushes CHAIR up while noise of the same magnitude leaves it near baseline. On Qwen, contrastive decoding is worse than magnitude-matched noise, which means its harm is a structured effect of the amateur branch, not generic perturbation.

On variance. The intervals above capture image-sampling variance for a single noise draw. They do not capture the variance of the random draw itself, which would require re-running each proxy several times (a GPU cost we leave for future work). As a rough check, two independent VCD runs on the same 500 LLaVA images gave CHAIR-S 53.0 and 55.0, a gap of about two points that is comparable to the difference between real VCD and its noise proxies. This is consistent with the

Table 3: d **at the wider top-30 window.** Mean of d over the expert's top-30 object candidates. On LLaVA the values are near zero and the real-versus- hallucinated ordering is not stable across windows; on Qwen hallucinated d remains clearly above real d.

		Top-30 mean d	
Model	Method	Real	Halluc.
LLaVA-1.5-7B	VCD	$+0.041$	-0.004
	SID	$+0.154$	-0.018
Qwen2.5-VL-7B	VCD	-0.917	-0.629
	SID	-0.721	-0.457

Table 4: $d = E - A$ **on emitted object words** (exact caption-level labels). In every setting the emitted hallucinated adjustment is higher than the real one, the opposite of suppression. For LLaVA-SID this reverses the sign seen in the candidate windows (Tables 2, 3), showing the LLaVA gaps are not a stable signal.

		Real objects		Hallucinated objects	
Model	Method	mean d	% $d>0$	mean d	% $d>0$
LLaVA-1.5-7B	VCD	$+0.264$	63.4	$+0.789$	74.0
	SID	$+0.296$	56.2	$+0.684$	63.1
Qwen2.5-VL-7B	VCD	-1.281	22.0	-0.097	40.7
	SID	-0.747	31.5	$+0.429$	48.2

reading that, on LLaVA, real contrastive decoding and matched noise are not distinguishable at the resolution of this experiment.

3.3 Observations

1. **No selective suppression of hallucinations.** On Qwen, hallucinated objects are consistently less suppressed than real objects across all three populations, the opposite of the method's stated goal. On LLaVA the adjustments are near zero and their sign is not stable across populations, so they provide no evidence of suppression. H1 is clearly rejected on Qwen and unsupported on LLaVA; the single cell consistent with H1 (LLaVA-SID candidates) is negligible in size and reverses under the exact-label view.
2. **The sign of the adjustment is model-specific.** Contrastive decoding amplifies object logits on LLaVA and suppresses them on Qwen. The one constant is that hallucinations are never suppressed more than real objects in a stable way.
3. **On LLaVA the method is indistinguishable from noise.** Real and proxy CHAIR intervals overlap on both metrics, so the amateur branch adds nothing a matched random perturbation does not.
4. **On Qwen the method is worse than noise.** Matched noise leaves CHAIR near greedy while the real subtraction raises it. H2 holds on Qwen, but not in the method's favour.
5. **Neither method beats greedy.** On LLaVA the real methods are within variance of greedy; on Qwen they are clearly above it.

3.4 Critical analysis

Strengths. The analysis targets the exact quantity the method adds to the expert score, $d = E - A$, rather than an indirect proxy for it. It triangulates across three populations (selection-free top-10 and top-30 candidates, and exact-label emitted words), which is what lets us see that the LLaVA effect is unstable rather than mistaking one window for a signal. The noise proxy is a like-for-like control that matches the measured magnitude of d and is tested in two forms (Gaussian and empirical bootstrap) that agree, and the CHAIR comparison now carries bootstrap intervals. Two models and two methods, on the same paired images, guard against a single-model artifact.

Table 5: **CHAIR for real methods versus noise proxies**, with image-level bootstrap 95% CIs ($B = 10{,}000$; percentages, lower is better). Proxy A adds Gaussian noise matched to the mean and standard deviation of d; Proxy B resamples the empirical d values. On LLaVA the proxy intervals overlap the real methods; on Qwen the proxies stay near greedy while the real methods rise.

Model	Setting	CHAIR-S	CHAIR-I
LLaVA-1.5-7B	Greedy	50.0 [45.6, 54.4]	13.9 [12.5, 15.3]
	VCD (real)	55.0 [50.6, 59.4]	15.6 [14.1, 17.0]
	SID (real)	54.2 [49.8, 58.6]	15.1 [13.7, 16.6]
	VCD noise proxy A	52.4 [48.0, 56.6]	14.8 [13.3, 16.3]
	VCD noise proxy B	52.8 [48.4, 57.2]	14.3 [12.9, 15.8]
	SID noise proxy A	49.6 [45.2, 53.8]	14.2 [12.6, 15.8]
	SID noise proxy B	50.0 [45.6, 54.2]	14.5 [12.9, 16.0]
Qwen2.5-VL-7B	Greedy	30.8 [26.8, 34.8]	7.7 [6.5, 8.9]
	VCD (real)	36.4 [32.4, 40.6]	10.5 [8.9, 12.1]
	SID (real)	38.4 [34.2, 42.6]	10.5 [9.1, 12.0]
	VCD noise proxy A	30.8 [26.8, 34.8]	8.2 [6.9, 9.5]
	VCD noise proxy B	31.0 [27.0, 35.2]	8.3 [7.0, 9.6]

Limitations, stated plainly. (1) The main proxy perturbs only object-category tokens, whereas the real methods perturb the whole vocabulary. We attempted a full-vocabulary control, but the only run available was on a 70-image subset, and in that subset the baseline relationship itself differs from the full sample (greedy scores above real VCD, the reverse of the 500-image result), so the subset cannot validate the proxy's fidelity. We therefore do not report it as evidence; a full-vocabulary proxy at 500 images, on both models, requires a GPU run and is the clearest remaining gap. (2) The proxy CHAIR intervals capture image-sampling variance only; noise-draw variance is estimated roughly from two independent runs but not measured properly, which again needs repeated GPU runs. (3) The values of d are read from stored top-30 lists, so a token absent from either list is skipped; this is rare and applies to both branches equally. (4) The candidate populations use single-token labels, with a residual multi-word case (for example "bear" from "teddy bear") affecting under one percent of candidates; the emitted population has no such caveat. (5) The noise parameters (Table 1) are computed over a broad pool of object tokens that differs from the reported d-analysis populations, which is why their mean does not equal any single table value; the exact pool is defined in Section 2.3. (6) The Qwen SID noise proxy was not run, so the proxy comparison covers three of the four model and method cells. (7) CHAIR ground truth is incomplete, so an object present but neither segmented nor named in any reference caption is scored as a hallucination; this applies identically to every run.

What a reviewer should take away. The contrastive adjustment is not a selective hallucination suppressor. On LLaVA its effect on CHAIR is indistinguishable from matched random noise, and on Qwen it is worse than noise, suppressing real objects more than hallucinations. This is a mechanistic account of why the reported gains do not materialise, and it complements the benchmark-level critique by locating the failure in the logit adjustment itself.

References

Shuai Bai, Keqin Chen, Xuejing Liu, et al. Qwen2.5-vl technical report. *arXiv preprint arXiv:2502.13923*, 2025.

Fushuo Huo, Wenchao Xu, Zhong Zhang, Haozhao Wang, Zhicheng Chen, and Peilin Zhao. Self-introspective decoding: Alleviating hallucinations for large vision-language models. *arXiv preprint arXiv:2408.02032*, 2024.

Sicong Leng, Hang Zhang, Guanzheng Chen, Xin Li, Shijian Lu, Chunyan Miao, and Lidong Bing. Mitigating object hallucinations in large vision-language models through visual contrastive decoding. In *CVPR*, 2024.

Tsung-Yi Lin, Michael Maire, Serge Belongie, James Hays, Pietro Perona, Deva Ramanan, Piotr Dollár, and C Lawrence Zitnick. Microsoft coco: Common objects in context. In *ECCV*, 2014.

Haotian Liu, Chunyuan Li, Yuheng Li, and Yong Jae Lee. Improved baselines with visual instruction tuning. In *CVPR*, 2024.

Anna Rohrbach, Lisa Anne Hendricks, Kaylee Burns, Trevor Darrell, and Kate Saenko. Object hallucination in image captioning. In *EMNLP*, 2018.

Hao Yin, Guangzong Si, and Zilei Wang. The mirage of performance gains: Why contrastive decoding fails to mitigate object hallucinations in multimodal large language models? *arXiv preprint arXiv:2504.10020*, 2025.